

How Far India Has Come In Building SUSTAINABLE CITIES

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In this article we look at some of the latest smart city solutions deployed across India

The very concept of smart cities is an amalgamation of city-level initiatives that drive ecological solutions to deliver a sustainable, comfortable and healthy living environment. Since the inception of the initiative in 2015, a good number of real-time smart city solutions have been deployed in the country. These are paving the way towards better waste management, tighter security, reduction of carbon footprint, better energy efficiency and more.

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Waste management solutions

India generates a massive volume of solid waste, amounting to more than 130 kilotonnes every day, of which, barely 25 kilotonnes get treated. Many Indian cities have been up in arms to mitigate these challenges. Ranging from smart bins to

robotic road cleaners, smart cities have been deploying interesting technologies to keep themselves clean.

For instance, New Delhi Municipal Council (NDMC) has installed reverse vending machines for waste collection that reward those who dispose plastic waste into the machines. The machines accept dry wastes like plastic cups, glasses, papers and plastic bags. Depending on the waste being submitted, citizens get a nominal financial reward. The reward

can be collected by downloading the required smartphone application, which has an e-wallet for the money to get credited.

The system can be used for other operations as well, such as ticket bookings, NDMC bill payments and even some Aadhaar-based services. Many such machines have been placed across Connaught Place and India Gate areas of New Delhi.

Surat in Gujarat has created an underground waste disposal channel to prevent open and untreated waste accumulation. Throughout the city, 43 underground garbage bins with overhead disposal inlets have been installed. These have a total waste holding capacity of 1.5 tonnes. The inlets have segregated openings for dry and wet waste.

Sensors fitted in the underground bins notify the waste control team when these detect that the bins are filled up to 70 per cent capacity. The team can then clear the bins in due time.

The process has massively boosted urban cleanliness, ranking Surat as one of the cleanest cities in the country. Surat treats 57 million litres of sewage daily, which gets converted to 40 million litres of potable water.

Jamshedpur in Jharkhand also employs interesting waste management facilities to boost their Zero Liquid Discharge project. Apart from real-time sensorised bins and wastewater treatment units, it has also set up biogas stations where wastes are converted into reusable biogas. Sewage drain covers are customised with sensors that detect when sewage fluid is about to overflow, so that teams can be deployed accordingly.

Renewable energy

Solar panels are the most convenient choice for renewable energy production. Commercial enterprises, government institutions as well as domestic households across the nation are installing solar setups to reduce the load on grid-based electricity. Collaterally, carbon gas emission also takes a step back.

Dry waste collection bin

